Nonlinear Preconditioning for Optimization and Monotone Inclusions

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Abstract

Many algorithms used in modern optimization, such as gradient clipping, sign-based and spectral optimizers like Muon, as well as adaptive schemes for (L_0, L_1)-smooth problems, can be understood through the lens of nonlinear preconditioning in the dual space. Here, the search direction is determined by passing the gradient, or more generally the operator, through the gradient of a Legendre convex reference function. In this talk I will give an overview of my group's recent work on this idea. I will start with minimization, where the theory is now well developed, thanks to tools of generalized convexity and duality [1, 2, 3, 4, 5, 6]. I will then move to monotone inclusion problems, where many basic questions are still open [7, 8].

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